

Dietary patterns and risk of pancreatic cancer in a large population-based case-control study in the San Francisco Bay Area

Pancreatic cancer is highly lethal, and identifying modifiable risk factors could have substantial public health impact. In this population-based case-control study (532 cases, 1701 controls), we used principal component analysis and multivariable unconditional logistic regression models to examine whether a particular dietary pattern was associated with risk of pancreatic cancer, adjusting for other known risk factors. A Prudent dietary pattern, characterized by greater intake of vegetables, fruit, fish, poultry, whole grains, and low-fat dairy, was associated with an approximate 50% reduction in pancreatic cancer risk among men (OR=0.51, 95% CI 0.31-0.84, p-trend=0.001) and women (OR=0.51, 95% CI 0.29-0.90, p-trend=0.04). A Western dietary pattern, characterized by higher intake of red and processed meats, potato chips, sugary beverages, sweets, high fat dairy, eggs, and refined grains, was associated with a 2.4-fold increased risk of pancreatic cancer among men (95% CI 1.3-4.2, p-trend=0.008); but was not associated with risk among women. Among men, those in the upper quintiles of the Western diet and lower quintiles of the Prudent diet had a 3-fold increased risk. Consistent with what has been recommended for several other chronic diseases, consuming a diet rich in plant-based foods, whole grains, and white meat, might reduce risk of pancreatic cancer.